



Graph Fractional Hilbert Transform, Analytic Signal, and Its Application to ECG Classification

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Abstract

This paper presents a unified framework for graph fractional analytic theory, extending classical analytic signal concepts to the graph fractional domain. Specifically, we introduce the graph fractional Hilbert transform, the graph fractional analytic signal framework based on the conjugate symmetry property of the graph fractional Fourier transform, and graph fractional amplitude modulation (AM) and frequency modulation (FM) concepts. We analyze fundamental properties including linearity, orthogonality preservation, and shift-invariance. The proposed framework is applied to electrocardiogram (ECG) signal classification, where graph fractional AM and FM features capture critical waveform characteristics, leading to improved classification performance compared to traditional methods.

Keywords Graph signal processing · Graph fractional Fourier transform · Hilbert transform · Analytic signal · ECG classification

1 Introduction

Graph signal processing (GSP) has emerged as a powerful signal processing paradigm in recent years, attracting significant attention from the research community [24, 27, 30]. The fundamental capability of GSP lies in processing data defined on graph structures, which finds extensive applications in diverse fields including social networks, sensor networks, and bioinformatics [10, 12, 14]. While traditional signal processing techniques encounter numerous challenges when applied to irregularly structured data, GSP can effectively capture complex inter-data relationships by leveraging the

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structural information inherent in graphs, thereby enhancing both the performance and effectiveness of signal analysis [13, 26].

The graph Fourier transform (GFT) represents one of the fundamental tools in GSP [27, 30], enabling frequency domain analysis of graph signals through eigendecomposition of either the graph adjacency matrix or the graph Laplacian matrix [28]. With the advancement of research, the graph fractional Fourier transform (GFrFT) has been proposed as a more flexible analytical tool [8, 9, 38, 40, 42]. By introducing fractional order parameters that can be adjusted according to specific signal processing requirements, the GFrFT provides enhanced flexibility and extends the scope of applications in graph signal analysis and processing [15, 25, 39, 41]. Recent advances have proposed unified theories [1], Hilbert-space formulations [44], and joint time-vertex extensions [2]. These contributions highlight the growing importance of fractional methods for graphs, but most of them emphasize transform definitions, spectral properties, or learning aspects rather than analytic theory.

Analytic theory occupies a central position in traditional signal processing [11, 20]. Essential tools such as the analytic signal, Hilbert transform, and modulation analysis serve crucial roles in time-frequency analysis, instantaneous signal characteristic extraction, and non-stationary signal processing [5, 7, 31]. The Hilbert transform enables the conversion of real signals into analytic signals, thereby facilitating instantaneous amplitude and frequency analysis [20]. Modulation analysis has been widely adopted for signal demodulation and feature extraction in communication systems [19, 33]. Additionally, analytic theory based on the fractional Fourier transform (FrFT) has been extensively investigated, providing robust theoretical foundations and powerful analytical methods for signal processing applications [17, 18, 21, 29, 32, 37, 43].

In the context of graph signal processing, analytic theory demonstrates equally significant importance. Venkitaraman et al. introduced the graph Hilbert transform, graph analytic signal, and graph modulation analysis [35, 36]. These developments not only inherit the advantages of traditional signal processing but also exhibit unique benefits tailored to graph-structured data. For instance, the graph Hilbert transform demonstrates particular effectiveness in highlighting anomalies and discontinuities in graph signals while precisely locating the nodes where these discontinuities occur [36]. However, despite the rapid development of fractional transforms on graphs [1, 2, 44], a systematic analytic theory analogous to the classical Hilbert and analytic signal framework is still missing in the graph fractional domain.

Building upon the conjugate symmetry properties exhibited by the GFrFT [6, 35, 38] and leveraging its inherent flexibility, this paper proposes the graph fractional Hilbert transform and establishes the corresponding graph fractional analytic signal framework. Furthermore, theoretical frameworks for graph fractional amplitude modulation (AM) and graph fractional frequency modulation (FM) are developed. In this way, our work complements and extends recent contributions on GFrFT [1, 2, 44] by filling a theoretical gap, thereby providing new perspectives and tools for graph signal analysis. The main contributions of this work are summarized as follows:

- We introduce the graph fractional Hilbert transform, which generalizes the classical Hilbert transform to graph signals in the fractional Fourier domain.

- We establish the graph fractional analytic signal framework, enabling the construction of unilateral spectra in the graph fractional domain.
- We develop graph fractional amplitude modulation (AM) and frequency modulation (FM) concepts, which provide enhanced flexibility for feature extraction through the fractional order parameter.

These theoretical contributions extend the analytic theoretical framework of graph signal processing and provide enhanced capabilities for practical applications.

The remainder of this paper is organized as follows: Section 2 introduces the graph fractional Fourier transform and classical analytic theory. Section 3 presents the definitions of the graph fractional Hilbert transform and graph fractional analytic signal, analyzes their fundamental properties, and introduces the concepts of graph fractional AM and FM. Section 4 demonstrates the application of graph fractional AM and FM for feature extraction from electrocardiogram signals in heart rhythm abnormality classification tasks. Section 5 concludes the paper and discusses future research directions.

2 Preliminaries

2.1 Graph Fourier Transform

Consider a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{A}\}$, where $\mathcal{V} = \{v_0, v_1, \dots, v_{N-1}\}$ with $|\mathcal{V}| = N$ represents a finite set of vertices. The set of edges is denoted by $\mathcal{E}$, and $\mathbf{A}$ is the weighted adjacency matrix. In this graph, if an edge $e = (i, j)$ connects vertices i and j , the weight of the edge is represented as $w_{i,j}$; otherwise, $w_{i,j} = 0$ [30]. The Jordan decomposition of adjacency matrix $\mathbf{A}$ is expressed as

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}, \quad (1)$$

where $\mathbf{\Lambda}$ is the Jordan blocks corresponding to the eigenvalues, and $\mathbf{U}$ represents the generalized eigenvectors.

A signal on the graph $\mathcal{G}$ can be described as [40]

$$\begin{aligned} f : \mathcal{V} &\rightarrow \mathbb{R} \\ v_i &\mapsto f(i). \end{aligned} \quad (2)$$

Alternatively, we can represent $\mathbf{f}$ as a real-valued vector $\mathbf{f} = [f(0), f(1), \dots, f(N-1)]^T \in \mathbb{R}^N$, where $f(i)$ denotes the value associated with the i -th vertex. Then, the GFT of $\mathbf{f}$ is defined as [30]

$$\hat{\mathbf{f}} = \mathcal{F}\mathbf{f} = \mathbf{U}^{-1}\mathbf{f}. \quad (3)$$

2.2 Graph Fractional Fourier Transform

Given a signal $f \in L^2(\mathbb{R})$, the fractional Fourier transform (FrFT) of f is given by [29]

$$\hat{f}_\alpha = \int_{\mathbb{R}} f(t) K_\alpha(t, u) dt, \quad (4)$$

where the kernel of FrFT is

$$K_\alpha(t, u) = \begin{cases} A_\alpha \exp \left\{ j \frac{u^2}{2} \cot(\alpha) - jtu \csc(\alpha) + j \frac{t^2}{2} \cot(\alpha) \right\} & \alpha \neq n\pi \\ \delta(u - t), & \alpha = 2n\pi \\ \delta(u + t), & \alpha = 2n\pi - \pi \end{cases}. \quad (5)$$

and $A_\alpha = \sqrt{\frac{1-j \cot(\alpha)}{2\pi}}$, $\alpha = \frac{p\pi}{2}$, $0 < |p| < 2$, $0 < |\alpha| < \pi$.

The GFrFT operator is defined as the fractional power of the GFT analysis operator $\mathcal{F} = \mathbf{U}^{-1}$:

$$\mathcal{F}_\alpha = \mathbf{P} \mathbf{J}^\alpha \mathbf{P}^{-1}, \quad (6)$$

where $\mathcal{F} = \mathbf{P} \mathbf{J} \mathbf{P}^{-1}$ is the Jordan decomposition of $\mathcal{F} = \mathbf{U}^{-1}$. The matrix $\mathbf{P}$ contains the (generalized) eigenvectors of $\mathcal{F}$ and is assumed to be invertible. The matrix $\mathbf{J}$ is block-diagonal, composed of Jordan blocks $\mathbf{J}_m(\gamma) = \gamma \mathbf{I} + \mathbf{N}$, with $\mathbf{N}$ nilpotent and upper-triangular, where m denotes the size of each Jordan block. Each block $\mathbf{J}_m(\gamma)$ admits a fractional power defined via binomial expansion $\mathbf{J}_m(\gamma)^\alpha = \sum_{r=0}^{m-1} \binom{\alpha}{r} \gamma^{\alpha-r} \mathbf{N}^r$. In the diagonalizable case, $\mathbf{J}^\alpha = \text{diag}(\gamma_0^\alpha, \gamma_1^\alpha, \dots, \gamma_{N-1}^\alpha)$.

For a graph signal $\mathbf{f}$, its GFrFT is defined as [38, 40]

$$\hat{\mathbf{f}}_\alpha = \mathcal{F}_\alpha \mathbf{f}, \quad (7)$$

where α denotes the fractional order. The inverse transform is given by

$$\mathbf{f} = \mathcal{F}_{-\alpha} \hat{\mathbf{f}}_\alpha = \mathbf{P} \mathbf{J}^{-\alpha} \mathbf{P}^{-1} \hat{\mathbf{f}}_\alpha. \quad (8)$$

2.3 The Discrete Hilbert Signal and Discrete Analytic Transform

Consider the classical Hilbert transform and analytic signals. Let $F(\omega)$ represent the discrete Fourier transform (DFT) of a real 1D signal $\mathbf{f}$. The discrete Hilbert transform $\mathbf{f}^{(h)}$ has the following frequency-domain specification: $F^{(h)}(\omega) = -i \text{sgn}(\omega) F(\omega)$, that is

$$F^{(h)}(\omega) = \begin{cases} -iF(\omega), & \omega \in \left\{ \frac{2\pi}{N}, \dots, \pi - \frac{2\pi}{N} \right\} \\ F(\omega), & \omega \in \{0, \pi\} \\ iF(\omega), & \omega \in \left\{ \pi + \frac{2\pi}{N}, \dots, \frac{2\pi(N-1)}{N} \right\} \end{cases}, \quad (9)$$

where $i = \sqrt{-1}$ and $\text{sgn}(\cdot)$ denotes the signum function [11, 36]. The discrete analytic signal of $\mathbf{f}$ in the time domain, defined as $\mathbf{f}^{(a)} = \mathbf{f} + i\mathbf{f}^{(h)}$, can also be represented in the frequency domain as $F^{(a)}(\omega) = (1 + \text{sgn}(\omega))F(\omega) = 2F(\omega)\mathcal{U}(\omega)$, where $\mathcal{U}(\cdot)$ denotes the unit-step function. Thus,

$$F^{(a)}(\omega) = \begin{cases} 2F(\omega), & \omega \in \left\{ \frac{2\pi}{N}, \dots, \pi - \frac{2\pi}{N} \right\} \\ F(\omega), & \omega \in \{0, \pi\} \\ 0, & \omega \in \left\{ \pi + \frac{2\pi}{N}, \dots, \frac{2\pi(N-1)}{N} \right\} \end{cases}. \quad (10)$$

3 Graph Fractional Hilbert Transform and Graph Fractional Analytic Signal

3.1 Definition

This section extends the classical Hilbert transform and analytic signal concepts to the graph fractional domain. For real-valued asymmetric adjacency matrices, it is observed that fractional eigenvalues and eigenvectors exhibit conjugate symmetry, as they are either real or form complex conjugate pairs. Leveraging this property, we define a graph fractional Hilbert transform that generates a unilateral spectrum, similar to the classical Hilbert transform. We begin by examining the GFrFT behavior on undirected and directed graphs to demonstrate the conjugate symmetry properties that underpin our definitions.

Example 1 Two examples are presented to demonstrate the conjugate symmetry properties of the GFrFT, which serve as the foundation of the proposed definitions.

- **Undirected graph case:** Consider a 64-vertex Swiss roll graph with adjacency matrix shown in Figure 1 (a). For a random graph signal (Figure 1 (b)), the GFrFT with parameter $\alpha = 0.79$ yields the spectrum in Figure 1 (c). Since the Swiss roll graph is undirected, the Jordan decomposition reduces to eigenvalue decomposition, resulting in real eigenvalues.
- **Directed graph case:** Consider a sparse directed graph with 9 vertices and random weights (adjacency matrix in Figure 1 (d)). For a random graph signal (Figure 1 (e)), the GFrFT with $\alpha = 0.79$ produces the spectrum in Figure 1 (f). Here, the fractional eigenvalues include complex conjugate pairs, which is crucial for our subsequent definitions.

The conjugate symmetry observed in Example 1 forms the theoretical foundation for our definitions. To formalize this concept and construct our graph fractional Hilbert transform, we establish the following mathematical framework:

- Assumption 1 The graph adjacency matrix $\mathbf{A}$ is real and asymmetric, containing at least one pair of complex conjugate fractional eigenvalues. This ensures the existence of conjugate symmetry in the fractional spectrum.
- Assumption 2 Let $\Theta = [\theta_0, \dots, \theta_{N-1}]^\top$ represent the phase angles of the fractional eigenvalues, ranging from 0 to 2π . Multiple eigenvalues with the same

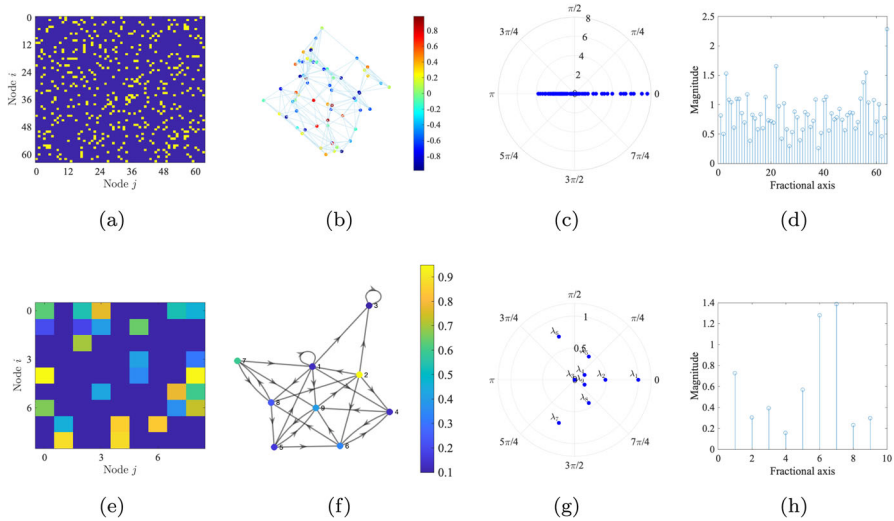


Fig. 1 Undirected graph case: (a) Adjacency matrix of the 64-vertex Swiss roll graph. (b) Random graph signal f . (c) Eigenvalues. (d) GFrFT with $\alpha = 0.79$. Directed graph case: (e) Adjacency matrix of the 9-vertex sparse directed graph. (f) Random graph signal g . (g) Eigenvalues. (h) GFrFT with $\alpha = 0.79$

phase angle are ordered by magnitude in descending order to ensure unique indexing.

Building upon these assumptions and inspired by the classical discrete Hilbert transform definitions in equations (9) and (10), a systematic approach is now established to construct unilateral fractional spectra. The central idea is to classify the fractional eigenvalues based on their phase angles, in a manner analogous to the classification of frequencies as positive or negative in the classical Hilbert transform. To implement this classification systematically, the different types of fractional eigenvalues are first counted. Let M_1 denote the number of real positive fractional eigenvalues, and M_2 denote the number of real negative fractional eigenvalues in Λ_α . The total number of real eigenvalues is given by $M = M_1 + M_2$.

The eigenvalue indices are divided into four distinct sets, each corresponding to a specific region in the complex plane:

$$\begin{aligned}\Gamma_1 &= \{1, 2, \dots, M_1\}, \\ \Gamma_2 &= \{M_1 + 1, M_1 + 2, \dots, M_1 + \frac{N - M}{2}\}, \\ \Gamma_3 &= \{M_1 + \frac{N - M}{2} + 1, M_1 + \frac{N - M}{2} + 2, \dots, \frac{N + M}{2}\}, \\ \Gamma_4 &= \{\frac{N + M}{2} + 1, \frac{N + M}{2} + 2, \dots, N\},\end{aligned}$$

where Γ_1 represents positive real-valued fractional eigenvalues, Γ_2 includes fractional eigenvalues with phase angles in $(0, \pi)$, Γ_3 contains negative real-valued fractional eigenvalues, and Γ_4 consists of fractional eigenvalues with phase angles in $(\pi, 2\pi)$.

This partition enables us to distinguish between "positive" and "negative" fractional frequencies in the graph domain, mirroring the classical frequency domain distinction.

The crucial property that enables our construction is the conjugate symmetry relationship between the sets Γ_2 and Γ_4 :

Property 1 For a real-valued graph signal $\mathbf{f}$, its GFrFT coefficients exhibit conjugate symmetry: for each $j \in \Gamma_2$, there exists a corresponding index $i \in \Gamma_4$ such that

$$\hat{f}_\alpha(j) = \hat{f}_\alpha^*(i), \quad j \in \Gamma_2, i \in \Gamma_4. \quad (11)$$

Specifically, for the case of all distinct fractional eigenvalues, we have

$$\hat{f}_\alpha(j) = \hat{f}_\alpha^*(N - j + M_1 + 1), \quad j \in \Gamma_2. \quad (12)$$

where $M_1 = |\Gamma_1|$ denotes the number of positive real fractional eigenvalues, and N is the total number of vertices.

Proof Let $\mathcal{F}_\alpha = [\mathbf{s}_0, \mathbf{s}_1, \dots, \mathbf{s}_{N-1}]$, where $\mathbf{s}_i$ correspond to the fractional eigenvectors. For real asymmetric adjacency matrices, the Jordan decomposition yields complex conjugate pairs. Following Assumption 2's ordering, for every j -th eigenvector such that $j \in \Gamma_2$, there exists an eigenvector indexed by $i \in \Gamma_4$ that shares a complex-conjugate relationship:

$$\mathbf{s}_j = \mathbf{s}_i^*, \quad j \in \Gamma_2, i \in \Gamma_4. \quad (13)$$

In particular, for the case of all distinct eigenvalues, we have:

$$\mathbf{s}_j = \mathbf{s}_{(N-j+M_1+1)}^*, \quad j \in \Gamma_2. \quad (14)$$

Then, as a consequence of the complex-conjugate relationship, for a real-valued graph signal $\mathbf{f}$, the GFrFT coefficients satisfy:

$$\hat{f}_\alpha(j) = \hat{f}_\alpha^*(i), \quad j \in \Gamma_2, i \in \Gamma_4. \quad (15)$$

This follows directly since $\hat{f}_\alpha(i) = \mathbf{s}_i^H \mathbf{f} = (\mathbf{s}_j^*)^H \mathbf{f} = (\mathbf{s}_j^H \mathbf{f})^* = \hat{f}_\alpha^*(j)$, where we use the fact that $\mathbf{f}$ is real-valued. $\square$

Let $\sigma = |\Gamma_1| + |\Gamma_2| + |\Gamma_3| = \frac{N+M}{2}$ denote the number of independent coefficients in the GFrFT representation. Property 1 establishes a fundamental conjugate symmetry relationship in the GFrFT matrix, which guarantees that for any real-valued graph signal $\mathbf{f}$, its GFrFT coefficients exhibit the relationship $\hat{f}_\alpha(j) = \hat{f}_\alpha^*(i)$ where $j \in \Gamma_2$ and $i \in \Gamma_4$. This symmetry has important implications for signal representation efficiency - only σ GFrFT coefficients are required to fully characterize a real graph signal, as the remaining coefficients can be uniquely determined through the conjugate symmetry property. This compact representation forms the theoretical basis for constructing the unilateral spectrum.

With this theoretical foundation established, we can now formally define the graph fractional Hilbert transform and graph fractional analytic signal.

Definition 1 For a graph signal $\mathbf{f}$ defined on the graph $\mathcal{G}$, its graph fractional Hilbert transform is defined as $\mathbf{f}_\alpha^{(gh)} = \mathcal{F}_{-\alpha} \hat{\mathbf{f}}_\alpha^{(gh)}$, where

$$\hat{f}_\alpha^{(gh)}(i) = \begin{cases} -i \hat{f}_\alpha(i), & i \in \Gamma_2 \\ 0, & i \in \Gamma_1 \cup \Gamma_3 \\ +i \hat{f}_\alpha(i), & i \in \Gamma_4 \end{cases} \quad (16)$$

Definition 2 For a graph signal $\mathbf{f}$ defined on the graph $\mathcal{G}$, its graph fractional analytic signal is defined as $\mathbf{f}_\alpha^{(ga)} = \mathcal{F}_{-\alpha} \hat{\mathbf{f}}_\alpha^{(ga)}$, where

$$\hat{f}_\alpha^{(ga)}(i) = \begin{cases} 2\hat{f}_\alpha(i), & i \in \Gamma_2 \\ \hat{f}_\alpha(i), & i \in \Gamma_1 \cup \Gamma_3 \\ 0, & i \in \Gamma_4 \end{cases} \quad (17)$$

Equivalently in the fractional spectral domain, $\hat{f}_\alpha^{(ga)}(i) = \hat{f}_\alpha(i) + i\hat{f}_\alpha^{(gh)}(i)$. The graph fractional analytic signal, as defined in (17), exhibits a unilateral spectral representation in the fractional domain. Through inverse GFrFT transformation, we obtain its vertex domain representation

$$\mathbf{f}_\alpha^{(ga)} = \mathbf{f} + i\mathbf{f}_\alpha^{(gh)}. \quad (18)$$

To validate our theoretical framework and demonstrate its practical utility, we present two comprehensive examples: Example 2 employs a synthetic social network graph to verify the unilateral spectrum property, demonstrating how our approach efficiently eliminates redundant negative frequency components while preserving the signal's essential characteristics. Example 4 explores the method's edge enhancement capabilities through application to image data, revealing how the graph fractional Hilbert transform effectively identifies signal discontinuities and highlights structural boundaries within the graph domain.

Example 2 Consider a synthetic social network graph with 5 distinct communities, where each community contains 3 member nodes. The connections among nodes are divided into two categories:

- Intra-community connections: Nodes within each community are "strongly connected," indicating that each node has a high likelihood of being connected to other nodes within the same community. The weights of these connections are drawn from a uniform distribution over the interval $[0,1]$, resulting in intra-community edge weights that can assume any value within this range.
- Inter-community connections: Nodes from different communities can also establish connections, albeit these connections are sparser compared to intra-community connections. The weights of inter-community connections are drawn from a uniform distribution over the interval $[0,0.5]$, making these connection weights generally smaller. These connections are randomly assigned, meaning that nodes from any two different communities have a potential to be connected, but the overall number of such edges is limited.

We define a dense inter-community link graph (Figure 2 (a)), in which the number of inter-community edges is relatively high, constituting 10% of the total possible edges. In this scenario, the graph exhibits a higher density of connections, leading to increased interaction between communities.

The adjacency matrix of this graph is shown in Figure 2 (b). Given the fractional parameter of the graph as $\alpha = 0.79$, the resulting fractional eigenvalues are depicted in Figure 2 (c). We then consider a graph signal with a unit amplitude in the GFrFT for each frequency, as shown in Figure 2 (d). The graph fractional Hilbert transform of this signal yields the GFrFT spectrum illustrated in Figure 2 (e), because $\Gamma_1 = \{1, 2, 3\}$, $\Gamma_2 = \{4, 5, 6\}$, $\Gamma_3 = \{7, 8, 9, 10, 11, 12\}$, $\Gamma_4 = \{13, 14, 15\}$. This corresponds to the graph fractional analytic signal with the GFrFT spectrum shown in Figure 2 (f). Figures 2 (g)-(i) display the vertex domain forms of the graph signals, the graph fractional Hilbert transform, and the graph fractional analytic signals, respectively.

Example 3 To assess computational efficiency, we measure the running time of the graph fractional Hilbert transform and the graph fractional analytic signal on synthetic social network graphs with sizes ranging from 50 to 1200 vertices. As shown in Figure 3, both methods exhibit similar super-linear growth, consistent with the $\mathcal{O}(N^3)$ preprocessing and $\mathcal{O}(N^2)$ per-signal cost of the GFrFT.

Example 4 Consider a 40×40 image, specifically a section of a coin image. Since image signals do not come with a predefined adjacency matrix, we define the graph as follows: each pixel located at position (i, j) is connected to the pixel at position $(i, j + 1)$ within the same row, as well as to the pixel at position $(i + 1, j)$ in the row below. Consequently, the graph's adjacency matrix is given by $\mathbf{A} = \mathbf{C} \otimes \mathbf{C}$, as illustrated in Figure 4 (a). Figures 4 (b) and (c) illustrate the original image signal and its corresponding graph fractional Hilbert transform, respectively. The application of the graph fractional Hilbert transform to this 2D image demonstrates edge enhancement capabilities. It is worth mentioning that variations in pixel connectivity lead to different directed graphs. While our experiments show that alternative graph constructions also highlight edges through the graph Hilbert transform, these cases are not elaborated here to prevent redundancy.

3.2 Properties and Discussions

For a given signal $\mathbf{f}$, we define $\mathbf{f}_\alpha^{(gh)} = \mathcal{H}_\alpha\{\mathbf{f}\}$ as its graph fractional Hilbert transform and $\mathbf{f}_\alpha^{(ga)} = \mathcal{A}_\alpha\{\mathbf{f}\}$ as its graph fractional analytic signal. The operator $\mathcal{A}_\alpha$ can be expressed as $\mathcal{A}_\alpha = \mathcal{I} + i\mathcal{H}_\alpha$, where $\mathcal{I}$ represents the identity operator. To characterize the graph fractional Hilbert transform, we introduce a diagonal matrix $\mathbf{\Lambda}_h$ with its i th diagonal element defined as

$$\mathbf{\Lambda}_h(i) = \begin{cases} -i, & \text{if } i \in \Gamma_2 \\ 0, & \text{if } i \in \Gamma_1 \cup \Gamma_3 \\ i, & \text{if } i \in \Gamma_4 \end{cases} \quad (19)$$

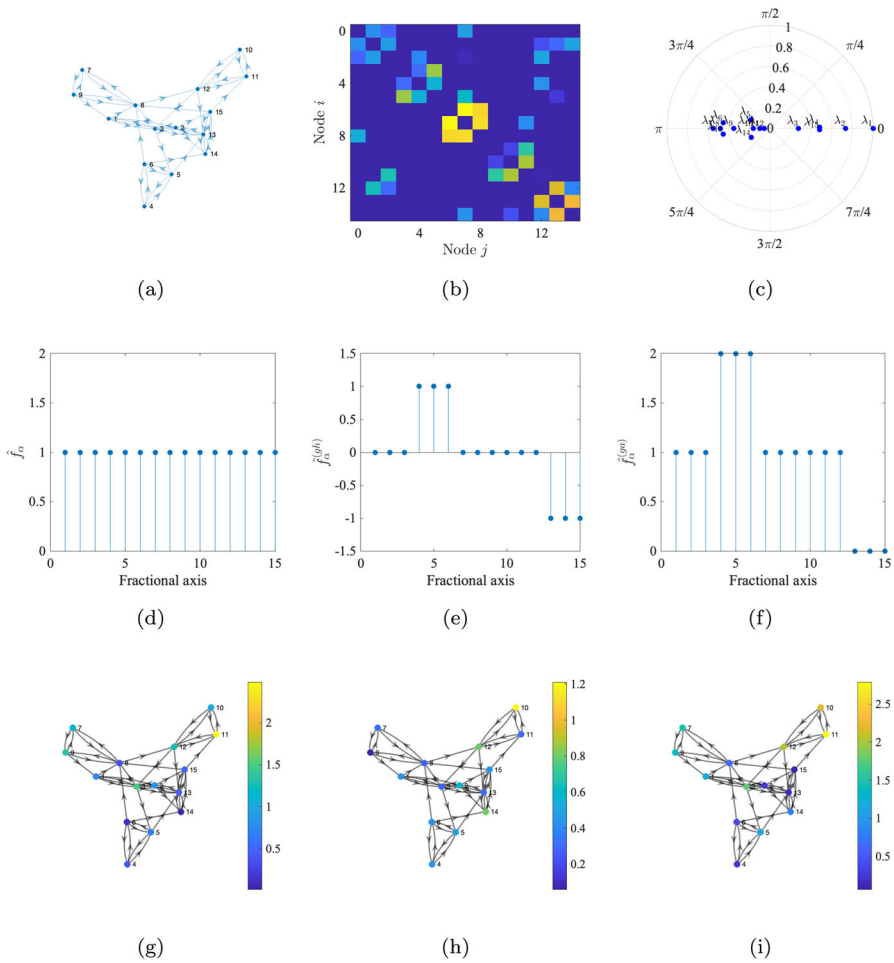


Fig. 2 (a) Dense inter-community link graph with 5 communities (3 nodes each). (b) Adjacency matrix. (c) Eigenvalues with $\alpha = 0.79$. (d) Graph signal with unit amplitude in GFrFT domain. (e) Graph fractional Hilbert transform. (f) Graph fractional analytic signal. (g)–(i) Vertex domain forms of (d)–(f), respectively

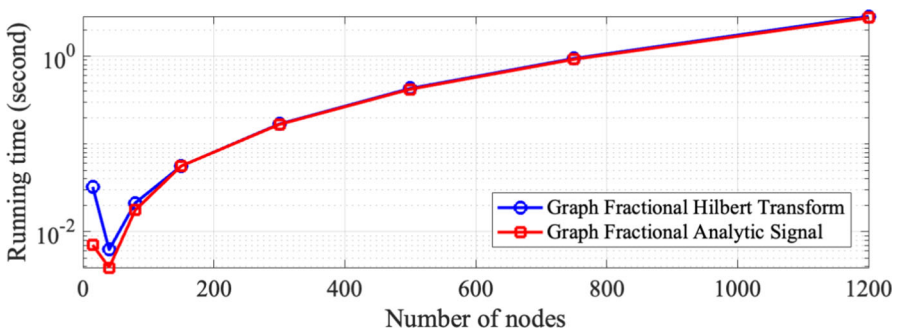


Fig. 3 Running time of the proposed methods on graphs of increasing size

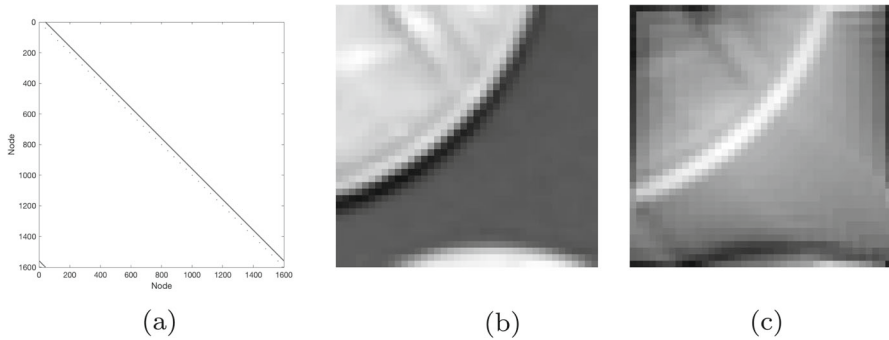


Fig. 4 Edge enhancement using the graph fractional Hilbert transform. (a) Adjacency matrix $\mathbf{A}$. (b) Original signal. (c) Graph fractional Hilbert transform with $\alpha = 0.7$

According to Definition 1, the graph fractional Hilbert transform operator can be formulated as $\mathcal{H}_\alpha\{\mathbf{f}\} = \mathcal{F}_{-\alpha}\mathbf{A}_h\mathcal{F}_\alpha\mathbf{f}$.

Next, some basic properties are presented.

Property 2 For $\mathbf{f}_1, \mathbf{f}_2 \in \mathbf{S}_2 \cup \mathbf{S}_4$ and $\beta_1, \beta_2 \in \mathbb{C}$, where $\mathbf{f} = \beta_1\mathbf{f}_1 + \beta_2\mathbf{f}_2$, we have

$$\mathcal{H}_\alpha\{\mathbf{f}\} = \beta_1\mathcal{H}_\alpha\{\mathbf{f}_1\} + \beta_2\mathcal{H}_\alpha\{\mathbf{f}_2\}. \quad (20)$$

Here, $\mathbf{S}_2 = \text{span}\{\mathbf{s}_i : i \in \Gamma_2\}$ and $\mathbf{S}_4 = \text{span}\{\mathbf{s}_i : i \in \Gamma_4\}$ denote the subspaces spanned by the fractional eigenvectors corresponding to Γ_2 and Γ_4 respectively.

Proof Since $\mathcal{H}_\alpha\{\mathbf{f}\} = \mathcal{F}_{-\alpha}\mathbf{A}_h\mathcal{F}_\alpha\mathbf{f}$, for $\mathbf{f} = \beta_1\mathbf{f}_1 + \beta_2\mathbf{f}_2$, we have

$$\begin{aligned} \mathcal{H}_\alpha\{\mathbf{f}\} &= \mathcal{F}_{-\alpha}\mathbf{A}_h\mathcal{F}_\alpha(\beta_1\mathbf{f}_1 + \beta_2\mathbf{f}_2) = \mathcal{F}_{-\alpha}\mathbf{A}_h(\beta_1\hat{\mathbf{f}}_{1,\alpha} + \beta_2\hat{\mathbf{f}}_{2,\alpha}) \\ &= \mathcal{F}_{-\alpha}(\beta_1\hat{\mathbf{f}}_\alpha^{(1,gh)} + \beta_2\hat{\mathbf{f}}_\alpha^{(2,gh)}) = \beta_1\mathcal{H}_\alpha\{\mathbf{f}_1\} + \beta_2\mathcal{H}_\alpha\{\mathbf{f}_2\}, \end{aligned}$$

where $\hat{\mathbf{f}}_\alpha^{(1,gh)}$ and $\hat{\mathbf{f}}_\alpha^{(2,gh)}$ denote the GFrFT of $\mathcal{H}_\alpha\{\mathbf{f}_1\}$ and $\mathcal{H}_\alpha\{\mathbf{f}_2\}$, respectively. $\square$

Property 3 The graph fractional Hilbert transform operator satisfies the following properties for signals in the subspace $\mathbf{S}_2 \cup \mathbf{S}_4$:

$$\mathcal{H}_\alpha^2 = -\mathcal{P}_{2,4}, \quad \text{where } \mathcal{P}_{2,4} \text{ is the projection onto } \mathbf{S}_2 \cup \mathbf{S}_4. \quad (21)$$

Furthermore, for $\mathbf{f} \in \mathbf{S}_2 \cup \mathbf{S}_4$, the repeated application of the operator results in

$$\mathcal{H}_\alpha^4\{\mathbf{f}\} = \mathbf{f}. \quad (22)$$

Proof From $\mathcal{H}_\alpha\{\mathbf{f}\} = \mathcal{F}_{-\alpha}\mathbf{A}_h\mathcal{F}_\alpha\mathbf{f}$, we obtain $\hat{\mathbf{f}}_\alpha^{(gh)} = \mathbf{A}_h\hat{\mathbf{f}}_\alpha$. Therefore, the GFrFT of $\mathcal{H}_\alpha^2\{\mathbf{f}\}$ is given by $\mathbf{A}_h^2\hat{\mathbf{f}}_\alpha$. Based on (19), we have

$$\mathbf{A}_h^2(i) = \mathbf{A}_h(i)\mathbf{A}_h(i) = \begin{cases} -1, & i \in \Gamma_2 \cup \Gamma_4 \\ 0, & i \in \Gamma_1 \cup \Gamma_3 \end{cases}. \quad (23)$$

This shows that Λ_h^2 acts as $-\mathcal{I}$ on the subspace corresponding to $\Gamma_2 \cup \Gamma_4$ and as the zero operator on the subspace corresponding to $\Gamma_1 \cup \Gamma_3$. Therefore, $\mathcal{H}_\alpha^2 = -\mathcal{P}_{2,4}$.

For signals $\mathbf{f} \in \mathbf{S}_2 \cup \mathbf{S}_4$, we have $\Lambda_h^4 \hat{\mathbf{f}}_\alpha = \hat{\mathbf{f}}_\alpha$ since $\Lambda_h^4(i) = 1$ for $i \in \Gamma_2 \cup \Gamma_4$, completing the proof. $\square$

Property 4 If $\mathcal{F}_\alpha$ is unitary and $\mathbf{f}_1, \mathbf{f}_2 \in \mathbf{S}_2 \cup \mathbf{S}_4$ are orthogonal, i.e., $\langle \mathbf{f}_1, \mathbf{f}_2 \rangle = 0$, then $\langle \mathcal{H}_\alpha \mathbf{f}_1, \mathcal{H}_\alpha \mathbf{f}_2 \rangle = 0$.

Proof Since $\mathbf{f}_1, \mathbf{f}_2 \in \mathbf{S}_2 \cup \mathbf{S}_4$ and $\mathcal{F}_\alpha$ is unitary, we have $\langle \hat{\mathbf{f}}_{1,\alpha}, \hat{\mathbf{f}}_{2,\alpha} \rangle = \langle \mathbf{f}_1, \mathbf{f}_2 \rangle = 0$. Then

$$\begin{aligned} \langle \mathcal{H}_\alpha \mathbf{f}_1, \mathcal{H}_\alpha \mathbf{f}_2 \rangle &= \langle \mathcal{F}_\alpha \mathcal{H}_\alpha \mathbf{f}_1, \mathcal{F}_\alpha \mathcal{H}_\alpha \mathbf{f}_2 \rangle = \langle \Lambda_h \hat{\mathbf{f}}_{1,\alpha}, \Lambda_h \hat{\mathbf{f}}_{2,\alpha} \rangle \\ &= \sum_{i \in \Gamma_2 \cup \Gamma_4} |\Lambda_h(i)|^2 \hat{f}_{1,\alpha}^*(i) \hat{f}_{2,\alpha}(i) = \sum_{i \in \Gamma_2 \cup \Gamma_4} \hat{f}_{1,\alpha}^*(i) \hat{f}_{2,\alpha}(i) \\ &= \langle \hat{\mathbf{f}}_{1,\alpha}, \hat{\mathbf{f}}_{2,\alpha} \rangle = \langle \mathbf{f}_1, \mathbf{f}_2 \rangle = 0, \end{aligned} \quad (24)$$

where we used the fact that $|\Lambda_h(i)|^2 = 1$ for $i \in \Gamma_2 \cup \Gamma_4$, $\Lambda_h(i) = 0$ for $i \in \Gamma_1 \cup \Gamma_3$, and that $\mathbf{f}_1, \mathbf{f}_2 \in \mathbf{S}_2 \cup \mathbf{S}_4$ implies their GFrFT coefficients are nonzero only for $i \in \Gamma_2 \cup \Gamma_4$. $\square$

Property 5 For graphs that can be diagonalized, the graph fractional Hilbert transform operates as a linear, shift-invariant graph filtering process.

Proof The graph fractional Hilbert transform can be expressed as $\mathcal{H}_\alpha\{\mathbf{f}\} = \mathcal{F}_{-\alpha} \Lambda_h \mathcal{F}_\alpha \mathbf{f} = \mathbf{H}_\alpha \mathbf{f}$, where $\mathbf{H}_\alpha = \mathcal{F}_{-\alpha} \Lambda_h \mathcal{F}_\alpha$.

Let $\mathbf{A}_\alpha := \mathcal{F}_{-\alpha} \Lambda \mathcal{F}_\alpha$ be the graph fractional shift operator. For a graph filter $\mathbf{K} = \sum_{i=0}^K k_i \mathbf{A}_\alpha^i = k(\mathbf{A}_\alpha)$ with $K \leq N$, the linearity and shift-invariance of $\mathbf{H}_\alpha$ can be demonstrated by showing that $\mathbf{H}_\alpha \mathbf{K} \mathbf{f} = \mathbf{K} \mathbf{H}_\alpha \mathbf{f}$. The filter $\mathbf{K}$ can be written as $\mathbf{K} = \mathcal{F}_{-\alpha} k(\Lambda) \mathcal{F}_\alpha$. For an input signal $\mathbf{f}$, the filter output $\mathbf{g} = \mathbf{K} \mathbf{f}$ in the GFrFT domain is given by

$$\hat{\mathbf{g}}_\alpha = \mathcal{F}_\alpha \mathbf{g} = k(\Lambda) \hat{\mathbf{f}}_\alpha. \quad (25)$$

Given that $\hat{\mathbf{f}}_\alpha^{(gh)} = \Lambda_h \hat{\mathbf{f}}_\alpha$, and using the commutativity of $k(\Lambda)$ and Λ_h , we obtain

$$\hat{\mathbf{g}}_\alpha^{(gh)} = \Lambda_h \hat{\mathbf{g}}_\alpha = \Lambda_h k(\Lambda) \hat{\mathbf{f}}_\alpha = k(\Lambda) \Lambda_h \hat{\mathbf{f}}_\alpha = k(\Lambda) \hat{\mathbf{f}}_\alpha^{(gh)}. \quad (26)$$

By applying the inverse GFrFT to equation (26), we conclude that

$$\mathbf{g}_\alpha^{(gh)} = \mathbf{H}_\alpha k(\mathbf{A}_\alpha) \mathbf{f} = k(\mathbf{A}_\alpha) \mathbf{f}_\alpha^{(gh)} = k(\mathbf{A}_\alpha) \mathbf{H}_\alpha \mathbf{f}, \quad (27)$$

which establishes the linearity and shift-invariance of the graph fractional Hilbert transform. $\square$

The Property 5 can in principle be extended to non-diagonalizable cases using the Jordan decomposition, but the resulting formulas are considerably more cumbersome and are therefore not pursued here.

Having established the fundamental properties of the graph fractional Hilbert transform, we now examine its computational complexity. Since the graph fractional Hilbert transform is defined as $\mathcal{H}_\alpha\{\mathbf{f}\} = \mathcal{F}_{-\alpha}\mathbf{A}_h\mathcal{F}_\alpha\mathbf{f}$, its computational cost is primarily determined by the GFrFT operations. For a graph signal $\mathbf{f}$ with N vertices, the GFrFT requires two successive Jordan decompositions: first decomposing the adjacency matrix $\mathbf{A}$ and then decomposing $\mathbf{U}^{-1}$, each requiring $\mathcal{O}(N^3)$ operations. This results in a total preprocessing cost of $\mathcal{O}(N^3)$, while each signal transformation involves forward and inverse GFrFT operations with $\mathcal{O}(N^2)$ complexity each. Therefore, the overall complexity is $\mathcal{O}(N^3)$ for initialization and $\mathcal{O}(N^2)$ per signal transformation.

3.3 Modulation Analysis

Building upon the established graph fractional Hilbert transform, we extend the classical modulation analysis framework to the graph fractional domain. In traditional signal processing, modulation analysis through AM and FM provides essential insights into signal structures and variations [22]. While classical signals evolve over time, graph signals vary across vertices where the topology encodes underlying relationships. This fundamental difference motivates our extension of modulation concepts to graph domains.

The graph fractional analytic signal, constructed using the graph fractional Hilbert transform, forms the foundation of our framework. Based on the definitions in Section 3.1, we now formally define this analytic signal and examine its key properties for modulation analysis.

Definition 3 For a graph signal $\mathbf{f}$ defined on graph $\mathcal{G}$, the graph fractional analytic signal is given by

$$\mathbf{f}_\alpha^{(ga)} = \mathbf{f} + i\mathbf{f}_\alpha^{(gh)} = \mathbf{f} + i\mathcal{H}_\alpha\{\mathbf{f}\}, \quad (28)$$

where $\mathbf{f}_\alpha^{(gh)} = \mathcal{H}_\alpha\{\mathbf{f}\}$ represents the graph fractional Hilbert transform of $\mathbf{f}$.

The graph fractional analytic signal inherits several important properties from its classical counterpart. Most notably, a unilateral spectrum is exhibited in the graph fractional domain, effectively eliminating redundant negative frequency components while preserving the complete signal information. This property is essential for subsequent modulation analysis, as it ensures that the extracted amplitude and frequency components convey meaningful signal characteristics without spectral aliasing effects.

The modulation analysis framework decomposes the graph fractional analytic signal into its amplitude and phase components, enabling separate analysis of signal magnitude variations and phase dynamics across the graph structure.

Definition 4 For a graph signal $\mathbf{f}$ with graph fractional analytic signal $\mathbf{f}_\alpha^{(ga)}$, the graph fractional AM and graph fractional instantaneous phase are defined as

$$\chi_\alpha(i) = \left| \mathbf{f}_\alpha^{(ga)}(i) \right| = \sqrt{[\mathbf{f}(i)]^2 + [\mathbf{f}_\alpha^{(gh)}(i)]^2}, \quad (29)$$

$$\phi_\alpha(i) = \arg \left(\mathbf{f}_\alpha^{(ga)}(i) \right) = \arctan \left(\frac{\mathbf{f}_\alpha^{(gh)}(i)}{\mathbf{f}(i)} \right), \quad (30)$$

for $i \in \{0, 1, \dots, N-1\}$, where $\arg(\cdot)$ denotes the argument function with values in $(-\pi, \pi]$.

The graph fractional AM $\chi_\alpha(i)$ quantifies the local signal intensity at each vertex i , incorporating both the original signal value and its Hilbert transform. The instantaneous phase $\phi_\alpha(i)$ captures the local phase behavior, which is essential for understanding signal oscillations and transitions in the graph domain. In 1D, the instantaneous frequency is obtained by the backward difference of the unwrapped phase. On graphs, we replace the time shift with a graph shift and keep the same residual form.

Definition 5 Let ϕ_α^u denote the unwrapped phase of $\mathbf{f}_\alpha^{(ga)}$, obtained by removing 2π discontinuities from ϕ_α . The graph fractional FM is defined as

$$v_\alpha(i) = \phi_\alpha^u(i) - \sum_{j=0}^{N-1} \mathbf{A}(i, j) \phi_\alpha^u(j), \quad (31)$$

where $\mathbf{A}(i, j)$ represents the (i, j) -th element of the adjacency matrix.

This definition generalizes the classical backward difference operator used in 1D frequency modulation to the graph domain. To illustrate this, consider a directed cycle graph with N vertices. Let $\mathbf{C}(i, j) = \begin{cases} 1, & j = (i-1) \bmod N, \\ 0, & \text{otherwise,} \end{cases}$ denote the backward-shift permutation matrix for all $i, j \in \{0, 1, \dots, N-1\}$. Substituting $\mathbf{C}$ into (31), the graph fractional FM becomes

$$v_\alpha(i) = \phi_\alpha^u(i) - \sum_{j=0}^{N-1} \mathbf{C}(i, j) \phi_\alpha^u(j) = \phi_\alpha^u(i) - \phi_\alpha^u((i-1) \bmod N), \quad (32)$$

which is exactly the circular (periodic) backward difference of the unwrapped phase. Similarly, for a directed path graph where the adjacency matrix $\mathbf{A}_{\text{path}}$ satisfies $\mathbf{A}_{\text{path}}(i, i-1) = 1$ for $i \geq 1$ and zero elsewhere, (31) reduces to

$$v_\alpha(i) = \phi_\alpha^u(i) - \phi_\alpha^u(i-1), \quad (33)$$

which matches the standard (non-circular) backward difference operator in 1D. The first vertex follows the chosen boundary condition. The graph fractional FM $v_\alpha(i)$ thus

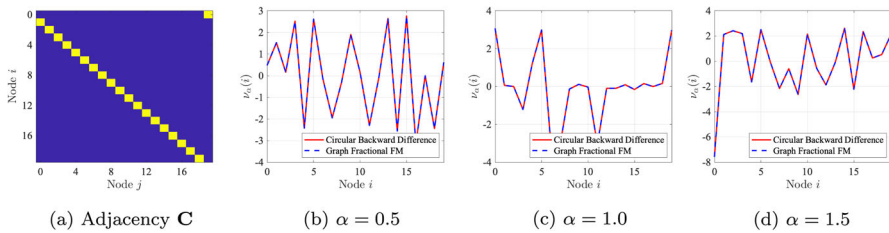


Fig. 5 Verification of (31) on a directed cycle. (a) Adjacency matrix $\mathbf{C}$ of the backward-shift operator. (b)–(d) Numerical comparison between ν_α from (31) and the circular backward difference applied to the same phase signal.

provides a unified way to express instantaneous frequency across both classical time series and graph-based domains, with consistent behavior under appropriate graph structures.

Example 5 We consider a directed cycle with $N = 20$ nodes and use the test signal $f(i) = \sin(2\pi(0.1i + 0.1i^2/2))$, $i \in 0, 1, \dots, N-1$, which exhibits increasing instantaneous frequency. The unwrapped phase ϕ_α^u is computed via the proposed graph fractional analytic construction. Applying (31) yields $\nu_\alpha(i)$ values that match the classical circular backward difference applied to the same phase. As shown in Figure 5, the two curves coincide across different values of α , confirming the consistency of the proposed formulation with classical 1D frequency modulation on directed cycles.

4 ECG Classification Based on the Graph Fractional AM and FM

Arrhythmia is a prevalent cardiovascular condition, often diagnosed using electrocardiograms (ECGs). With advancements in science and technology, real-time ECG diagnosis using mobile electronic devices has emerged as a significant research area. Automatic computer-aided classification of arrhythmias is a key focus within this field [4, 16].

We propose a novel classification method for ECG signals utilizing the graph fractional AM and graph fractional FM derived from our graph fractional Hilbert transform framework. While traditional graph Hilbert transform and analytic signal methods have been explored, our approach extends these concepts to the fractional domain, providing enhanced flexibility through the fractional order parameter α . This fractional extension enables more adaptive feature extraction, which can be optimized for specific signal characteristics. To validate this approach, the proposed graph fractional AM/FM method is compared with traditional AM/FM baseline methods.

The MIT-BIH Arrhythmia Database, available at <https://archive.physionet.org/cgi-bin/atm/ATM>, is utilized in this study. This database provides comprehensive data across a wide range of categories and is annotated by authoritative experts, making it widely adopted in arrhythmia research [3, 23]. Since abnormal waveform signals associated with general arrhythmias typically manifest within a single heartbeat, the complete electrocardiogram sequences from the MIT-BIH database are first segmented

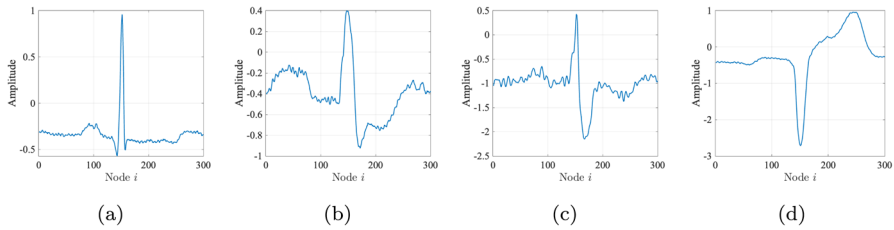


Fig. 6 The waveforms of the four types of heartbeats. (a) N. (b) LBBB. (c) RBBB. (d) PVC.

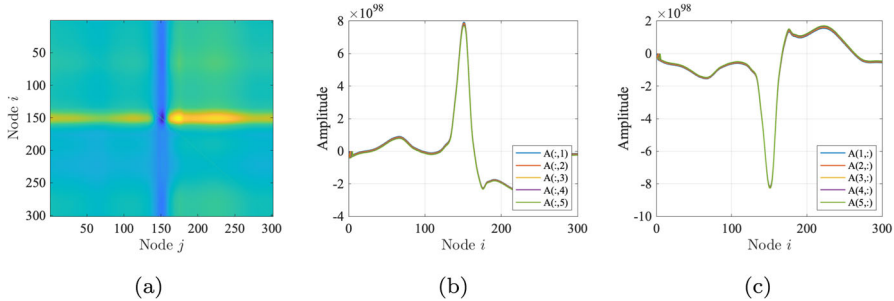


Fig. 7 (a) Adjacency matrix $\mathbf{A}^*$ obtained from (34). (b) First 5 rows. (c) First 5 columns.

into individual heartbeats. Using the R-wave position as a reference point, 150 sample points before and 150 sample points after the R-wave are extracted, resulting in 301 sample points per heartbeat. The analysis primarily focuses on normal beats (N) and three common types of arrhythmia: left bundle branch block beats (LBBB), right bundle branch block beats (RBBB), and premature ventricular contractions (PVC). The waveforms of the four heartbeat types are illustrated in Figure 6. For each type of heartbeat, 1000 samples are selected and denoted as $\mathbf{X}_N$, $\mathbf{X}_L$, $\mathbf{X}_R$, and $\mathbf{X}_P$, respectively.

To represent the electrocardiogram signal as a graph signal, it is necessary to generate the corresponding adjacency matrix. Let $\mathbf{X} = [\mathbf{X}_N; \mathbf{X}_L; \mathbf{X}_R; \mathbf{X}_P]$ (total number 4000), we determine the adjacency matrix by solving the following problem [34, 36]

$$\begin{aligned} \mathbf{A}^* &= \operatorname{argmin}_{\mathbf{A}} \|\mathbf{X} - \mathbf{A}\mathbf{X}\|_2^2 \\ \text{subject to } & \operatorname{diag}(\mathbf{A}) = \mathbf{0}, \mathbf{A}\mathbf{1} = \mathbf{1}, \mathbf{A}^\top \mathbf{1} = \mathbf{1}. \end{aligned} \quad (34)$$

To prevent ill-conditioning due to insufficient learning data, we impose the optimization condition $\mathbf{A}\mathbf{1} = \mathbf{1}$ and $\mathbf{A}^\top \mathbf{1} = \mathbf{1}$ in (34). This condition improves numerical stability and does not affect the generality of the proposed GFrFT-based AM/FM framework, which applies to arbitrary graph shifts. We solve the optimization problem in (34) using a gradient descent-based solver with a convergence tolerance of 10^{-6} and a maximum of 5000 iterations. The resulting adjacency matrix is shown in Figure 7(a), and Figures 7(b) and (c) present its first five rows and columns, respectively.

Based on the obtained adjacency matrix $\mathbf{A}^*$, we compute the graph fractional Hilbert transform and graph fractional AM for the four heartbeats shown in Figure 6, with

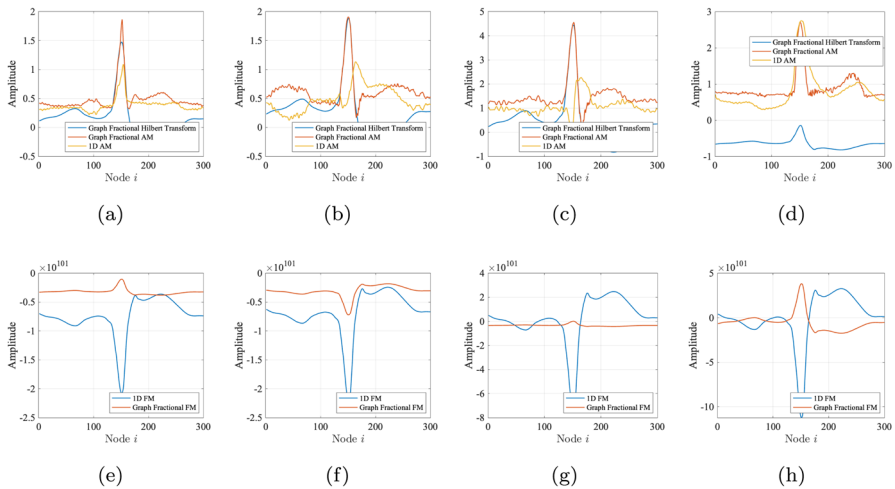


Fig. 8 Graph fractional Hilbert transform, graph fractional AM ($\alpha = 0.95$), and 1D AM: (a) N, (b) LBBB, (c) RBBB, (d) PVC. Graph fractional FM and 1D FM: (e) N, (f) LBBB, (g) RBBB, (h) PVC.

the results presented in Figure 8. Figures 8(a)-(d) display the graph fractional Hilbert transform and graph fractional AM with $\alpha = 0.95$, as well as the 1D AM for the four different heartbeats. Figures 8(e)-(h) show the graph fractional FM and 1D FM for the four different heartbeats. The graph fractional AM and FM effectively capture the features of the ECG signals. Note that Figure 8 displays graph-domain quantities indexed by vertices. Since the learned graph is not constrained to preserve temporal order, the vertex index is not a time axis, and clinical time-domain annotations are not directly aligned without an explicit node-time mapping.

A two-layer neural network classifier is employed, as illustrated in Figure 9. The input layer consists of 602 nodes, followed by two hidden layers containing 11 and 4 neurons, respectively, with the hyperbolic tangent activation function. The output layer contains 4 nodes. The maximum number of training steps is set to 1000, and the learning rate is fixed at 0.05.

To demonstrate the effectiveness of our graph fractional Hilbert transform framework, we compare two classification approaches:

- **Classifier 1:** This classifier utilizes the DFT of traditional 1D AM and 1D FM as features of the ECG signal.
- **Classifier 2:** This classifier employs the GFrFT of the proposed graph fractional AM and graph fractional FM as features of the ECG signal.

Note that Classifier 2 naturally reduces to the conventional graph Hilbert transform framework when $\alpha = 1$, i.e., the GFrFT degenerates to the standard GFT-based formulation. Therefore, in the subsequent parameter analysis experiments (e.g., Figure 8(b)), the case $\alpha = 1$ directly corresponds to the traditional GFT baseline, while $\alpha \neq 1$ illustrates the effect of the fractional extension.

For the dataset containing 4000 samples, 70% are randomly selected as the training set, while the remaining 30% are used as the testing set. Here, accuracy is defined as

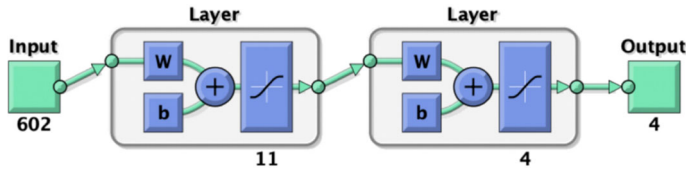


Fig. 9 Architecture of the two-layer neural network classifier.

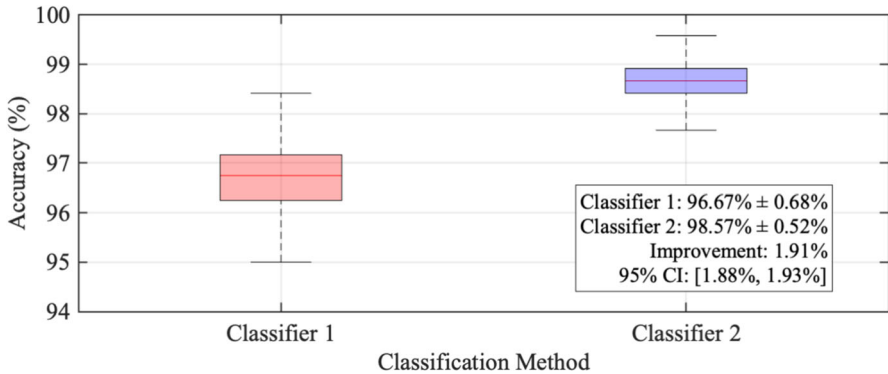


Fig. 10 Comparison of classification accuracies obtained by Classifier 1 and Classifier 2.

the ratio of correctly classified samples to the total number of samples. The fractional order parameter used in Classifier 2 is set to $\alpha = 0.95$. We conduct 10,000 experiments on the dataset to ensure the robustness of the results. As shown in Figure 10, Classifier 1 achieves an average accuracy of $96.67\% \pm 0.68\%$, while Classifier 2 reaches $98.57\% \pm 0.52\%$, yielding an absolute improvement of 1.91%. The 95% confidence interval of [1.88%, 1.93%] indicates that this gain is highly consistent across trials. To assess its statistical significance, we perform a paired t -test, which produces $t = 172.35$ with $p < 0.001$, decisively rejecting the null hypothesis of equal performance. This demonstrates that the proposed method provides a statistically significant improvement, and the observed gain is unlikely to be attributed to random variation. This improvement demonstrates the effectiveness of our graph fractional Hilbert transform and analytic signal framework in extracting more informative features from ECG signals. The fractional extension provides additional flexibility that enables the graph fractional AM and FM to capture richer signal characteristics compared to traditional methods.

Finally, two key aspects of the graph fractional framework are investigated. Example 1 focuses on the optimization of the neural network architecture, while Example 2 explores the impact of the fractional order parameter α on the performance of the graph fractional AM and FM features. The parameter analysis highlights the flexibility inherent in the fractional extension of graph analytic signal methods. To obtain more stable classifier performance results, the accuracy is averaged over 10 experiments for each group.

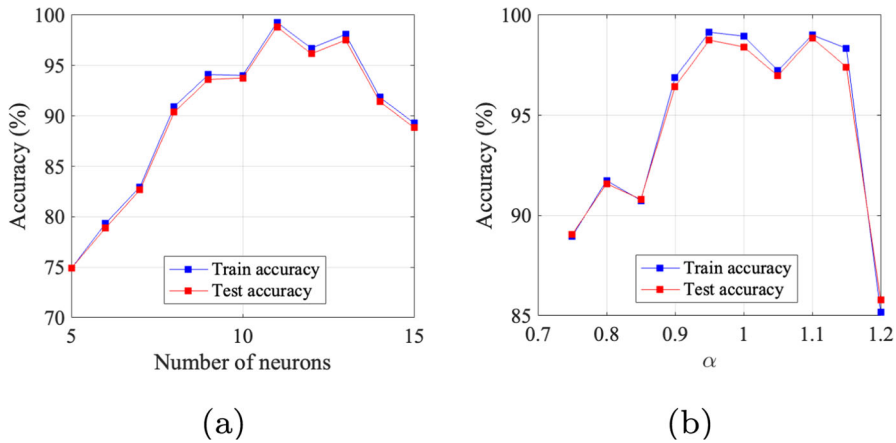


Fig. 11 Classification accuracy: (a) with varying numbers of neurons, (b) with different fractional parameters

Example 6 The number of neurons in the first layer is varied from 5 to 14, with a consistent fractional order parameter of 0.95. The classifier performance results are shown in Figure 11(a). Optimal performance is achieved with 11 neurons, as using either too few or too many neurons leads to a decline in accuracy.

Example 7 The fractional order parameter α is systematically varied from 0.75 to 1.2 to examine its impact on the graph fractional AM and FM features. The results shown in Figure 11(b) indicate that optimal performance on both the training and testing sets is achieved at $\alpha = 0.95$. Notably, when $\alpha = 1$, the method naturally reduces to the traditional graph AM and FM framework, which serves as the baseline. This analysis highlights a key advantage of extending graph analytic signal methods to the fractional domain: the parameter α introduces additional flexibility, enabling the graph fractional Hilbert transform framework to adapt to different signal characteristics and potentially extract more discriminative features for various ECG patterns.

Through systematic analysis of the fractional order parameter α and neural network architecture, we have demonstrated the effectiveness of our novel graph fractional Hilbert transform framework for ECG signal classification. By extending traditional graph analytic signal theory to the fractional domain, our approach leverages graph fractional AM and FM features to achieve significantly better classification performance compared to conventional methods. The enhanced flexibility provided by the fractional framework enables better feature extraction and signal characterization, validating its potential for broader applications in graph signal processing. However, it is worth noting that the current parameter optimization approach is relatively basic and could benefit from further research. More sophisticated optimization methods, such as adaptive algorithms or meta-heuristic approaches, could potentially lead to better parameter selection and improved performance. This limitation presents an opportunity for future work to develop more advanced parameter optimization strategies.

5 Conclusion

This paper introduces the graph fractional Hilbert transform and graph fractional analytic signal, constructed by leveraging the conjugate symmetry properties of the graph fractional Fourier transform. The proposed framework extends classical analytic tools such as the Hilbert transform and analytic signal to the graph fractional domain, and establishes their key properties including linearity, orthogonality preservation, and shift-invariance. Furthermore, we formulated graph fractional amplitude modulation (AM) and frequency modulation (FM), thereby enriching the analytic theory of graph signal processing. Experimental validation on ECG classification demonstrates that the proposed features achieve higher accuracy compared to traditional approaches, showing that the fractional extension provides additional flexibility and stronger discriminative capability.

Nonetheless, several limitations remain. The need for Jordan decomposition brings high computational cost, which may restrict scalability to very large graphs. The method also relies on accurate graph construction and is sensitive to the choice of fractional order parameter, which may reduce robustness in real-time or noisy scenarios. Future work will focus on improving scalability and robustness through more efficient algorithms, adaptive parameter selection, and better graph modeling. To reduce computational complexity, we will also investigate scalable approximations such as Chebyshev polynomials and randomized low-rank methods. Moreover, the framework will be validated on noisier databases and a wider range of arrhythmia classes to assess its generalizability.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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